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3 August 1962

MILITARY SPECIFICATION

QUINONEDIOXIME, COMMERCIAL GRADE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements and tests for one type of commercial grade quinonedioxime.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

* PPP-D-705	Drum, Shipping and Storage, Steel, 16 and 30 Gallon Capacity
PPP-D-723	Drums, Fiber
* PPP-D-729	Drums, Shipping and Storage, Steel, 55 Gallon (208 Liter)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-ED, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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*	PPP-T-76	Tape, Packaging, Paper (For Carton Sealing)
*	RR-S-366	Sieve, Test
*	UU-P-268	Paper, Kraft, Wrapping

STANDARDS

* Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

* CODE OF FEDERAL REGULATIONS

Title 49(172.316) - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

* (Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, 25, DC).

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3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), the contractor shall furnish preproduction samples for preproduction inspection in accordance with 4.3. The preproduction samples shall be produced by the same methods, processes and techniques to be employed in subsequent production of the contract quantity.

3.2 Composition. The nominal composition of the quinonedioxime shall consist essentially of the chemical compound para-quinonedioxime. The maximum allowable impurities shall not exceed the amounts specified in Table I. The physical properties of the compound shall conform to the requirements specified in Table II.

TABLE I ALLOWABLE IMPURITIES

	Minimum	Maximum
Acetone insoluble, percent....		4.0
Ash, percent.....		1.0
Quinonemonoxime, percent.....	4.8	6.2
Volatile matter, percent.....		0.9

TABLE II. PHYSICAL PROPERTIES

Decomposition temperature, degrees centigrade (C)..	223
Particle size, percent retained on No. 200 US Standard sieve	0.2

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3.3 Workmanship. The quinonedioxime shall be a homogenous crystalline composition and shall be free of lumps, dirt, trash, contaminants and other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot size. A lot of quinonedioxime shall consist of that quantity produced, by one manufacturer, in not more than 24 consecutive hours of continuous production, with no change in facilities, materials, process or specifications which would change the chemical or physical properties of the composition. In the event the process is a batch operation, each batch shall constitute a lot.

4.3 Preproduction provisons.

4.3.1 Sample. The preproduction sample shall be that quantity of quinonedioxime specified by the procuring activity (see 6.2)

4.3.2 Inspection location. Preproduction inspection shall be performed by the Government at a place specified by the procuring activity (see 6.2).

4.3.3 Inspection. The sample specified in 4.3.1 shall be subjected to the chemical and physical tests specified in 4.5.

4.4 Production provisions.

4.4.1 Sampling.

4.4.1.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 at inspection level II.

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4.4.1.2 For chemical and physical tests.

4.4.1.2 Primary samples. Ten percent of the containers comprising each lot submitted shall be selected at random; except when lots comprise less than 100 containers, 10 or all the containers, whichever is less, shall be selected. A representative one-pint sample from each selected container shall be taken in the following manner: Insert a closed sampler tube to maximum depth, at an oblique angle, then open and agitate to fill the tube. Each sample shall be placed in a clean, dry, glass jar, which shall be labeled to identify the container and lot number.

4.4.1.2.2 Composite samples. The primary samples shall be divided into two groups of equal size, and a composite sample shall be prepared from each group. Each composite sample shall be placed in a clean, dry container, sealed, and labeled in a manner that will identify the sample with the containers and lot represented. Each composite sample shall be subjected to all chemical and physical tests specified in 4.5, and the average value of the results obtained from testing the two composite samples shall be taken as representative of the lot.

4.5 Tests.

4.5.1 Chemical Tests.

4.5.1.1 Acetone insoluble.

Reagent:	Anhydrous acetone, reagent grade.
Apparatus:	Filtering crucible, fine porosity, Selas No. 3001, or equal, prepared with an asbestos mat and heated for one hour at 500 degrees C.

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The determination shall be made as follows: Weigh accurately about one gram of the composite sample and transfer to a 250-milliliter (ml) beaker. Add 100 ml of acetone to the sample, heat on a steam bath, and stir intermittently until the mixture reaches a gentle boil. Cool the solution to room temperature and filter through the Selas filtering crucible. Wash free of soluble material with acetone. Save the filtrate for the quinonemonoxime determination. Dry the crucible and contents for one hour at 70 degrees C, cool, and weigh. Save the contents for the ash determination. Calculate the percent acetone insoluble as follows:

$$\text{Acetone insoluble, percent} = \frac{(A - B)\ 100}{W}$$

Where,

A = Weight of crucible and acetone insoluble,

B = Weight of crucible,

W = Weight of sample,

4.5.1.2 Ash. The determination shall be made as follows: Heat the crucible containing the precipitate from the acetone insoluble test to 500 degrees C for 3 hours in a muffle furnace, cool to room temperature, and weigh. Calculate percent of ash as follows:

$$\text{Ash, percent} = \frac{(C - B)\ 100}{W}$$

Where,

B = Weight of crucible,

C = Weight of crucible and ash,

W = Weight of sample,

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4.5.1.3 Quinonemonoxime.

Reagents: Anhydrous acetone, reagent grade.

Acetic acid solution, reagent grade, one percent by volume in acetone.

Activated alumina (see 6.3).

Methanol solution, 5 percent by volume in acetone.

Sodium hydroxide solution, 1.0 Normal (N).

Filter aid (see 6.3).

The determination shall be made as follows: Transfer the filrate from the acetone insoluble determination to a 250-ml beaker, evaporate to about 50 ml on a steam bath, and cool to room temperature. Prepare the absorption column with a glass wool plug at the constricted section. Add activated alumina in increments of 2 to 3 cm and pack each increment lightly by tapping the column on a rubber pad and tamping the alumina with a dowel. After the column is packed to within 2 inches of the top, center a 3/4 inch filter paper over the alumina and seat it against the alumina with the dowel. Transfer the sample to the column. Use 10 ml of acetone to complete the transfer. When the sample and wash acetone drain to the level of the packing, immediately wash with a 10 ml portion of acetic acid solution. When the acid solution has drained to the packing start washing with 300 ml of the methanol solution from a 500 ml separatory funnel, with the tip about one inch above the alumina packing. When the washing is complete, dry the column with suction for one-half hour. Remove the brown top layer and discard. Remove the green-yellow quininemenoxime layer quantitatively and transfer to a 250 ml beaker. Add approximately 0.8 gram of filter aid and wash the alumina with eight 50 ml portions of 1N sodium hydroxide solution to remove the quinonemonoxime, decanting each portion through a medium-porosity Selas crucible, or equal. (Analysis must be completed from this point without delay). Wash with two 25 ml portions of distilled water. Transfer the filtrate quantitatively to a one liter volumetric flask and make up to volume with distilled water. To redilute the solution to a concentration such that the

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absorbency is between 0.4 and 1.0, pipette and aliquot (usually 5 ml) into a 100 ml volumetric flask, add 6 ml of 1N sodium hydroxide solution, and dilute to volume with distilled water. Determine the absorbency (A) of a portion of this diluted solution, using distilled water as a blank, at 399 millimicrons with a slit width of 0.10 millimeter (mm) and 363 millimicrons with a slit width of 0.15 mm. Calculate the percent quinonemonoxime as follows:

$$\text{Quinonemonoxime, percent} = \frac{(5.25A - 1.08B)\ 10}{GV}$$

Where,

V = volume of the aliquot used, ml,

G = sample weight, gr,

A = A. at 399 millimicrons (quinonemonoxime peak),

B = A. at 363 millimicrons (quinonedioxime peak).

4.5.1.4 Volatile matter. The determination shall be made as follows: Weigh a sample of 2 to 3 grams in a tared low-form, glass-stoppered weighing bottle (about 50 mm). Place in a 70 degree C oven for 90, plus or minus 3, minutes, cool in a desiccator over desiccant, and weigh. Calculate percent volatile matter as follows:

$$\text{Volatile matter, percent} = \frac{(A - B)\ 100}{(A - C)}$$

Where,

A = weight of the weighing bottle and sample,

B = weight of the weighing bottle and dried sample,

C = weight of dry weighing bottle.

4.5.2 Physical property tests

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4.5.2.1 Decomposition temperature. The test for decomposition temperature shall be conducted as follows:

Apparatus: Melting point apparatus (see 6.3),

Melting point capillary tubes, 100 mm by 1.5 to 2.0 mm,

Variable transformer.

4.5.2.1.1 Calibration. Select at least four pure, powdered materials with melting points that cover the range of the thermometer to be calibrated. Prepare 3 melting-point tubes with each material. Select a transformer setting at which the temperature rises no more than 1/2 degree C per minute at the melting point of each material and record the 3 observed melting points of each material. Plot the observed melting points (y) against the known melting points (x) of the standard materials. If any material has an excessive melting-point range or deviates obviously from the line of best fit, purify and rerun the material or alternately, substitute another material with a similar melting point. Calculate the equation. Y = ax plus b, of the line of best fit by the method of least squares. The constants a and b for the apparatus are to be recorded along with the variable transformer settings for use with the given thermometer and apparatus.

4.5.2.1.2 Procedure. Collect a small amount of the composite sample in the open end of a capillary melting-point tube. Drop the capillary onto a hard surface through a section of glass tubing about one foot long, and repeat this process until approximately 2 cm of the capillary is packed with sample. (Do not seal the top end of the capillary.) Preheat the bath to approximately 100 degrees C. Attach the capillary to the calibrated thermometer, with the sample opposite the mercury bulb, and lower the assembly into the melting-point apparatus until the thermometer bulb is below the side-arm. The liquid of the bath shall be near the zero degree mark when decomposition occurs. Set the variable transformer control so that the temperature rise is approximately 8 degrees C per minute when decomposition occurs. The decomposition of quinonedioxime should occur with a small "puff" and liberation brown fumes.

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Record the observed decompositon point and calculate the true decomposition point as follows:

$$\text{Decomposition temperature, degrees C} = \frac{(y - b)}{a}$$

Where,

y = observed decomposition temperature, degrees C,

a and b = constants of calibration equation.

As an alternate, read the true value from a plot of the calibration curve.

4.5.2.2 Particle size. The particle size shall be determined as follows:

Apparatus: US Standard testing sieve, 8 inch diameter, size 200 conforming to RR-S-366.

2-inch paint brush, stiff bristles.

Transfer 100 grams of sample to a No. 200 US Standard testing sieve thoroughly wet with distilled water. Wash all the fines through the sieve with a flow of distilled water, brushing as necessary. Dry the sieve at 70 degrees C, carefully brush the retained material onto a paper, and weigh to 0.01 gram.

Calculation:

$$\text{Percent retained} = \frac{(\text{weight of residue})\ 100}{\text{weight of sample}}$$

4.6 Packing and marking. Packing and marking shall be inspected to determine compliance with the requirements of section 5.

* 5. PACKAGING

* 5.1 Preservation, packing, and unitization. Level A or B (see 6.2

5.1.1 Level A. The quinonedioxime shall be packed in drums, 25 and 100 pound net capacity, as specified in the contract or order.

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The 25 pound quantities shall be packed in drums conforming to PPP-D-705, type VI. The 100 pound quantities shall be packed in drums conforming to PPP-D-729, type IV. The drums shall be fabricated from steel not lighter than 26 gage for 25 pound capacity, and not lighter than 23 gage for 100 pound capacity. The drums shall have a bolted ring-type cover.

* 5.1.2 Level B. The quinonedioxime shall be packed directly in water-resistant fiberboard or plywood drums conforming to PPP-D-723, type II, grade A, and PPP-D-723 respectively. Net capacity of the drums shall be 25, 50, or 100 pounds as specified in the contract or order. Each plywood container shall be lined with a kraft paper bag fabricated from material conforming to UU-P-268. After filling, the ears or flaps of the liner shall be secured in place with tape conforming to PPP-T-76. The top-closure and strapping shall be in conformance to the applicable container specification.

5.2 Marking.

* 5.2.1 Normal marking. The exterior shipping containers shall be marked in accordance with the requirements of MIL-STD-129 and paragraph 172.316, Title 49, Code of Federal Regulations.

5.2.2 Special marking. In addition to any special marking required by the contractor or order, digit marking of exterior containers for overseas shipment shall be accomplished as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. Quinonedioxime is used as an ingredient in rocket solid propellants.

6.2 Ordering data. Procurement documents should specify the following:

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(a) Title, number and date of this specification.

(b) Whether a preproduction sample is required (see 3.1 and 4.3.1) and size of sample (see 4.31).

(c) Preproduction inspection location (see 4.3.2).

(d) Selection of applicable level of packing (see 5.1).

(e) Marking requirements if other than specified (see 5.2.1).

(f) Whether digit marking is required (see 5.2.2).

6.3 Definitions.

6.3.1 The following test equipment and laboratory supplies have been successfully used in complying with the provisions of this specification:

(a) Activated alumina, grade F20, Aluminum Company of America, or equal.

(b) "Hyflo Super Cel", Johns Manville Company, or equal.

(c) Spectrophotometer, Model DU, with matched Corex cells, set up for use with a tungsten lamp, ultraviolet-sensitive phototube, and red-purple filter, Beckman Instrument Company, or equal.

(d) Melting-point apparatus, catalog number J150, No. 23-100, filled with cottonseed oil "Arochler" Central Scientific Company or equal.

6.4 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in the notations. Bidders and contractors

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DOCUMENT IDENTIFIER AND TITLE
QUINONEDIOXIME, COMMERCIAL GRADE, MIL-Q-51093A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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